INFINITE PRODUCTS OF SURJECTIVE AFFINE MORPHISMS ARE SURJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. The existence of the product T is the input lemma lemma-infinite-product (existence asserted in the chapter with an omitted hint proof); this card's own content, the surjectivity, is fully worked in span.

Allowed standard prerequisites (editorial).

- Morphism from a point. Schemes, morphisms from spectra of fields

Source reference. Original lemma 0CNI: sources/stacks-chow/tags.txt, source lines 15893–15893. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let I be a set and for each $i \in I$ let $f_i : T_i \to S$ be a surjective affine morphism. Then the product $T = \prod T_i$ in the category of schemes over S (Lemma 0CNI) maps surjectively to S.*

Proof. Let $s \in S$. Choose $t_i \in T_i$ mapping to s. Choose a huge field extension $K/\kappa(s)$ such that $\kappa(s_i)$ embeds into K for each i. Then we get morphisms $\mathrm{Spec}(K) \to T_i$ with image s_i agreeing as morphisms to S. Whence a morphism $\mathrm{Spec}(K) \to T$ which proves there is a point of T mapping to s. □

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